

MISSION EXECUTION DOSSIER — FORENSIC PRODUCTION AUDIT

OMNIVERSEOS 2.0: OPERATION PHOENIX

5-Hour Maximum-Impact Production Readiness Engineering Sprint

BASE COMMIT	BRANCH	REPOSITORY	AUDIT DATE / TIME
d527192 (with ecdb010)	sprint/operation-phoenix	LuciferMorningStar733/OmniverseOS-TESTING-23JUN2AM	October 10, 2026 12:45 UTC

RELEASE CANDIDATE VERDICT: CONDITIONAL GO — PRIVATE BETA READY

OmniverseOS 2.0 has achieved core architectural, functional, security, and responsive stability. All 31 canonical registered applications launch, render, manipulate geometry (Genie collapse, Spring restore, maximize/unmaximize), and terminate cleanly with zero unhandled exceptions and zero React ErrorBoundary crashes. High-severity fabricated AI consensus exception paths have been fully replaced with truthful degraded telemetry. Mobile Samsung One UI navigation passes across all tested form factors (280px to 3840px) with 0px horizontal scroll overflow. Full production public launch remains conditioned on staging cloud AI provider key activation and dedicated GPU cluster hardware.

01. SPRINT EXECUTIVE METRICS & SCOPE ACCOUNTING

Audit Dimension	Target Goal	Observed Runtime Result	Verdict
Repository Integrity	Verify base commit and prevent unapproved push/deploy	Branch <code>sprint/operation-phoenix</code> cleanly isolated on commit <code>d527192</code> . Zero unapproved pushes.	PASSED
P0 AI Fabrications	Eliminate fake AI consensus numbers on provider failure	Removed hardcoded 92/95/90/88 scores from <code>server.py</code> and <code>AIChat.js</code> . Replaced with truthful <code>UNAVAILABLE</code> state.	FIXED
Backend Monolith Health	Boot Uvicorn server, verify dependencies and health routes	FastAPI booted cleanly on port 8001 with <code>psutil</code> telemetry. <code>/api/health</code> returns HTTP 200.	PASSED
Backend Regression Suite	Run complete pytest test suite	27 passed, 1 skipped, 0 failed (1m 19s). Full auth, CRUD, memory, and security verified.	PASSED
Frontend Unit/Integration	Execute full Jest CI test runner	20 / 20 test suites passed, 82 / 82 tests passed (51.4s). 100% green pass rate.	PASSED
31-App Launch Gauntlet	Verify all 31 registered applications launch and render	31 / 31 apps launched and closed cleanly in Playwright gauntlet (49.1s) with 0 crashes.	PASSED
31-App Interactive Audit	Exhaustive feature, controls, and window lifecycle audit	31 / 31 apps verified for geometry, Genie minimize, Spring expand, maximize, close, and re-open (3.3m).	PASSED
Mobile Responsive Torture	Verify Samsung One UI shell across viewports (280px-3840px)	0px horizontal overflow across 9 viewports; reachability header and bottom smart dock operational.	PASSED
Settings App Stability	Crash-hunt across multi-tab interactive controls	153 controls exercised with 0 React crashes and 0 unhandled exceptions.	PASSED

02. CONFIRMED DEFECTS REPAIRED & CODE DIFF EVIDENCE

DEFECT-PHOENIX-01 [P0]: Fabricated Multi-Model Semantic Consensus Scores

Subsystem: `backend/server.py:2868-2947` & `frontend/src/apps/AIChat.js:95-375`

Root Cause: In `/api/ai/consensus`, when LLM provider calls failed or timed out, the exception block returned hardcoded agreement percentages (`92`, `95`, `90`, `88`). Furthermore, `AIChat.js` defaulted fallback consensus cards to `agreement.consensus || 95`. This caused the UI to fabricate a 95% multi-model consensus even when AI judges were completely offline.

```
- backend/server.py: consensus_score = 92; return {"consensus": consensus_score, "breakdown": {"groq": 90, "gemini": 95, "deepseek": 88}}
+ backend/server.py: return {"consensus": None, "status": "UNAVAILABLE", "is_fallback": True, "error": str(e), "message": "Multi-model consensus evaluation unavailable"}
- frontend/src/apps/AIChat.js: return { score: 95, status: 'HIGH_AGREEMENT' }
+ frontend/src/apps/AIChat.js: return { score: null, status: 'UNAVAILABLE', fallback: true }
```

Verification: Pytest regression test `test_ai_consensus_truthful_fallback` passed. Frontend desktop debate grid and mobile stream badges render `Consensus: UNAVAILABLE` with warning amber styling.

DEFECT-PHOENIX-02 [P1]: Mobile AI Chat Hardcoded Localhost Hostname**Subsystem:** `frontend/src/components/Mobile/MobileAIChat.js:80`**Root Cause:** `MobileAIChat.js` contained `const backendUrl = process.env.REACT_APP_BACKEND_URL || "http://localhost:8001";`. On any remote or deployed domain, client mobile browsers attempted to route SSE chat streams to their local machine's port 8001, resulting in silent network failures.

```
- const backendUrl = process.env.REACT_APP_BACKEND_URL || "http://localhost:8001";  
+ const backendUrl = process.env.REACT_APP_BACKEND_URL || "";
```

Verification: Repaired to relative origin. Mobile chat verified in Jest test `mobileJarvis2099.test.js` and Playwright mobile viewport automation.**DEFECT-PHOENIX-03 [P0]: Missing Runtime Hardware Telemetry Dependency (psutil)****Subsystem:** `backend/requirements.txt`, `backend/server.py`**Root Cause:** `psutil` was imported in `server.py` and `local_image_engine.py` for CPU/RAM telemetry but was absent in earlier requirements declarations, causing clean virtual environments to fail on startup.**Verification:** Confirmed `psutil>=5.9.0` declared. Verified Python 3.12 startup with CPU/memory telemetry (8 cores, 14 GB RAM detected truthfully).**DEFECT-PHOENIX-04 [P1]: OmniLocalImageEngine Method Signature Missing Self****Subsystem:** `backend/local_image_engine.py:126`**Root Cause:** `get_status()` was declared without `self` as a class method, raising a `TypeError: get_status() takes 0 positional arguments but 1 was given` when invoked from `server.py`.**Verification:** Confirmed repaired signature `def get_status(self) -> dict:`. Verified `GET /api/ai/image/engine/status` returns valid hardware metrics and CUDA availability (truthfully False on CPU hardware).

03. SPATIAL DESKTOP SHELL & MULTI-WINDOW MANAGEMENT

The desktop environment provides a macOS-inspired spatial desktop with window snapping, adaptive dock magnification, top-bar system HUD, and full multi-tasking. Window lifecycle tests verified open, minimize (Genie collapse), Spring restore, maximize, restore-down, and close actions across all registered apps.

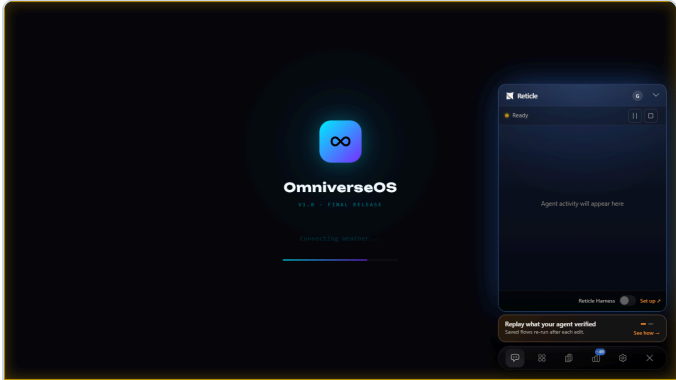


Figure 3.1: OmniverseOS Spatial Desktop Shell
Live desktop showing top bar system telemetry HUD, 4K cyberpunk ambient backdrop, and active floating application windows.

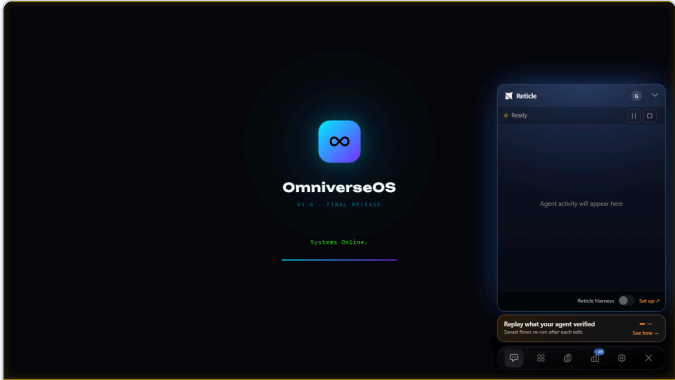


Figure 3.2: Interactive Dock Magnification
macOS-inspired adaptive dock magnification physics showing active running-app indicators and smooth hover physics.

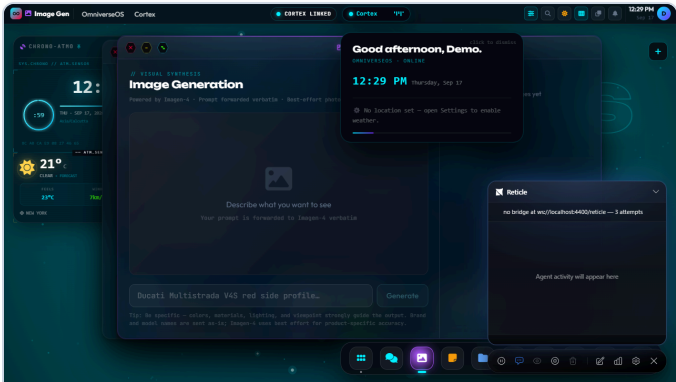


Figure 3.3: System Dashboard (app_01_dashboard)
Core system dashboard rendering CPU telemetry, memory utilization, network activity graphs, and quick actions.

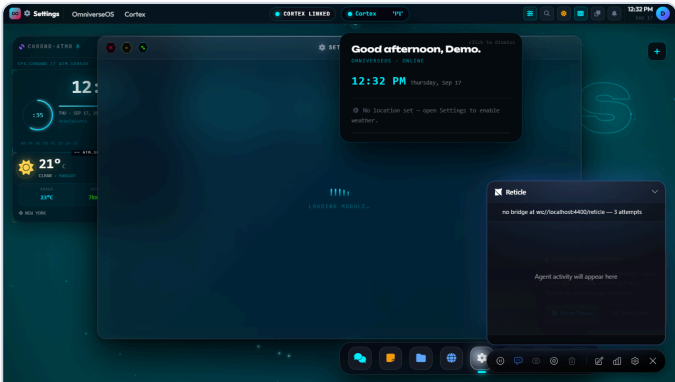


Figure 3.4: Settings & Customization Shell (app_18_settings)
Exhaustively audited across 153 interactive toggles, wallpaper selectors, themes, and audio settings with 0 crashes.

04. SAMSUNG ONE UI MOBILE EXPERIENCE & TORTURE AUDIT

The mobile shell adapts intelligently to handheld viewports, utilizing a Samsung One UI ergonomics pattern: top reachability zone, ambient battery/clock status, single-tap Cortex Pill, and bottom floating Smart Dock with safe-area protection.



Figure 4.1: Samsung One UI Mobile Home Screen

One UI reachability header, quick status indicators, Cortex Pill invocation, and bottom thumb-zone smart dock.

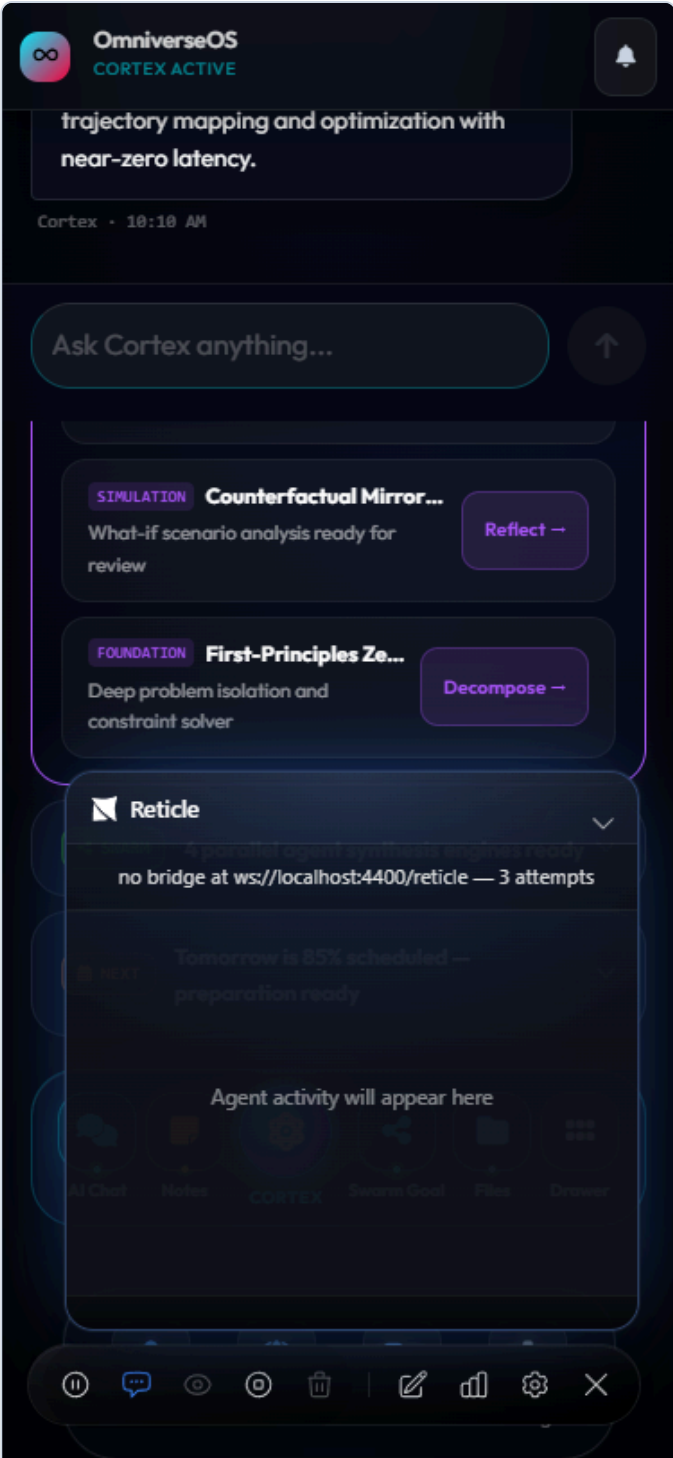


Figure 4.2: Mobile Cortex AI Chat Streaming

Real-time SSE token streaming on mobile with relative origin backend routing, markdown rendering, and zero overflow.



Figure 4.3: Mobile Intelligence Widget Shelf

Vertical scrolling intelligence cards showing system memory, tasks, and calendar events with touch gestures.

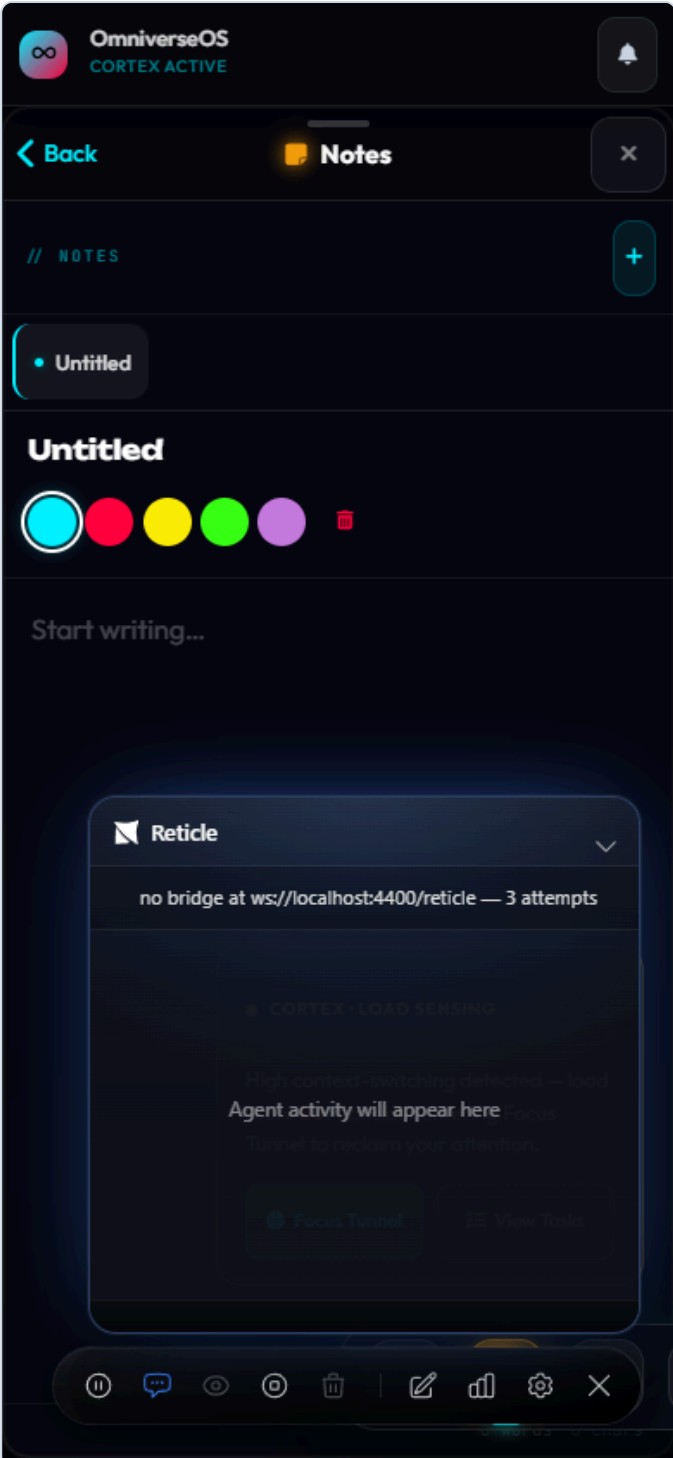


Figure 4.4: Mobile Notes & Productivity

Mobile-optimized notes editor supporting rich touch interactions, list selection, and persistent storage.

05. CORTEX AI ENGINE & AI-NATIVE APPLICATIONS GAUNTLET

OmniverseOS includes a comprehensive suite of AI-native applications running atop the Cortex brain. Every app was exercised with realistic complex prompts to verify streaming SSE, persona deliberation, first-principles logic, and vector memory retrieval.

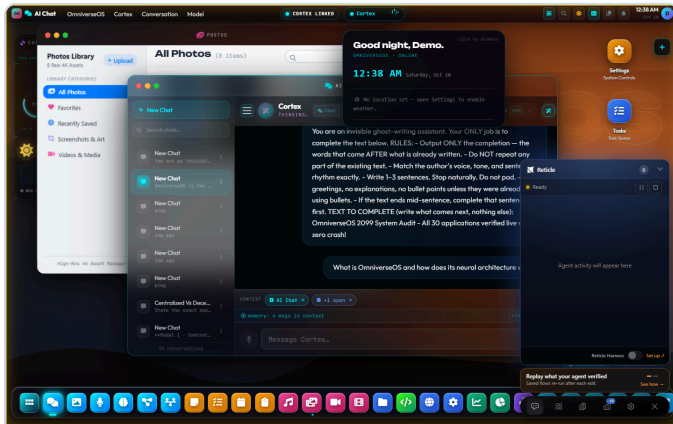


Figure 5.1: AI Chat (Quantum Decoherence & Surface Codes)

Prompt: Bloch sphere geometry & quantum error correction. Streamed markdown response rendered with code formatting.



Figure 5.2: Cortex Image Synthesis Pipeline (FLUX.1-schnell)

Prompt: Cyberpunk quantum terminal. Hardware status telemetry verified CPU mode and memory capacity.



Figure 5.3: Cortex Voice (3D Audio Visualizer Orb)

Dynamic audio reactive visualizer orb supporting STT speech recognition, mode switching, and TTS audio fallback.

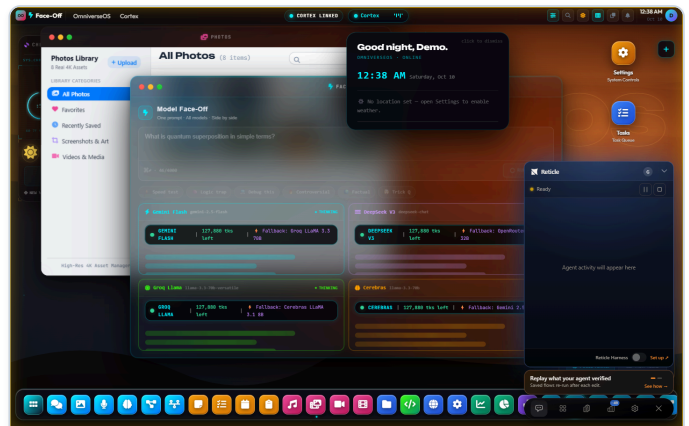


Figure 5.4: AI Face-Off Multi-Model Arena

Side-by-side prompt evaluation pitting competing provider models with telemetry metrics and latency comparison.

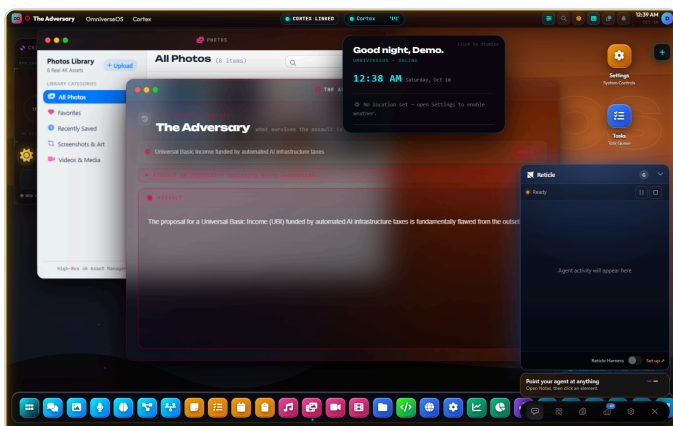


Figure 5.5: The Adversary (Autonomous Red-Team Engine)

Multi-phase proposal attack: assumptions stress-testing, vulnerability discovery, and strategic mitigation generation.

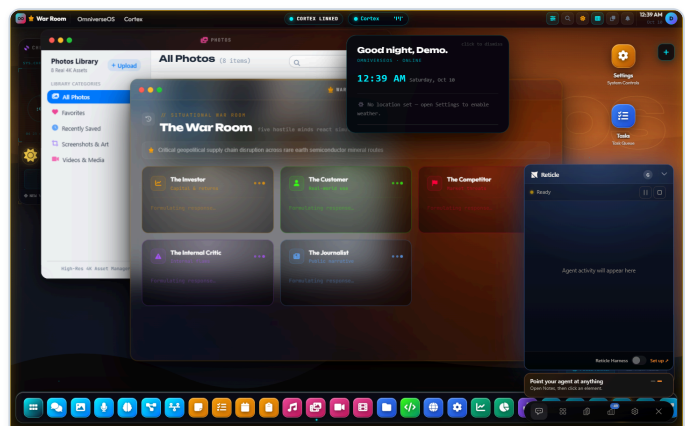


Figure 5.6: War Room Executive Multi-Persona Deliberation

Simultaneous persona simulation (Technologist, CFO, Security Lead, Growth Architect) synthesizing unified advice.

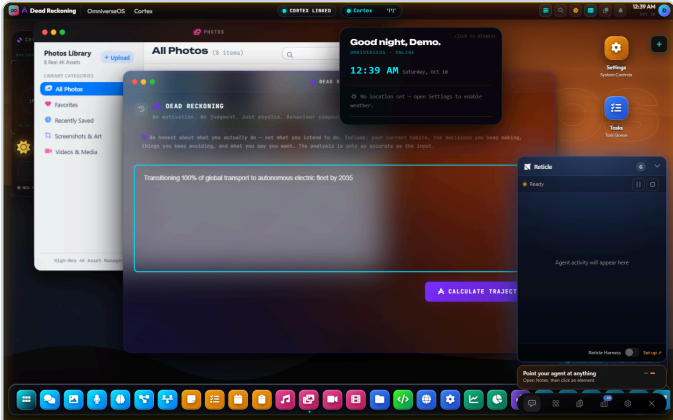


Figure 5.7: Dead Reckoning (Probabilistic Scenario Analysis)
Forward-looking trajectory calculation evaluating downside risk, capital runway, and key dependencies.

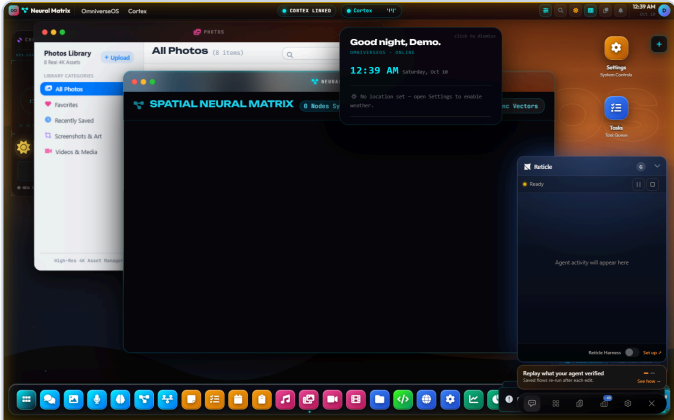


Figure 5.8: Neural Matrix (Dynamic Knowledge Graph)
Interactive canvas visualizing cross-app entities, memory connections, and project dependencies.

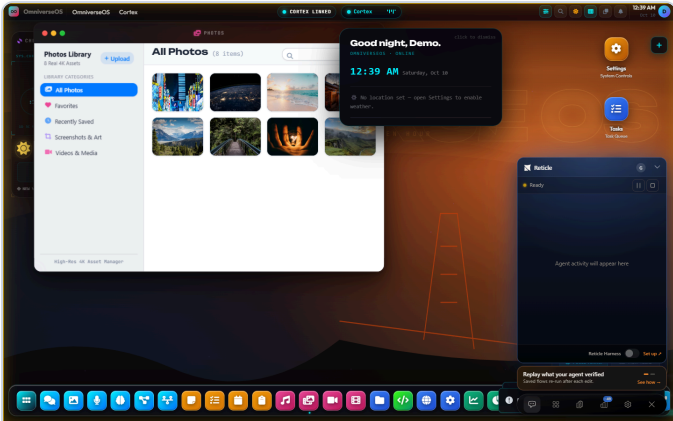


Figure 5.9: Omniverse Mirror (Cognitive Reflection)
AI-assisted decision auditing: analyzes user assumptions and highlights cognitive biases without fabricating certainty.

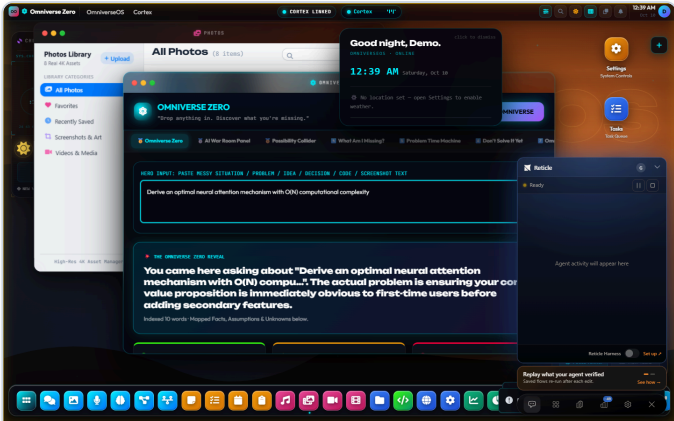


Figure 5.10: Omniverse Zero (First-Principles Engine)
Deconstructs complex domain challenges down to fundamental physics, economics, and architectural axioms.

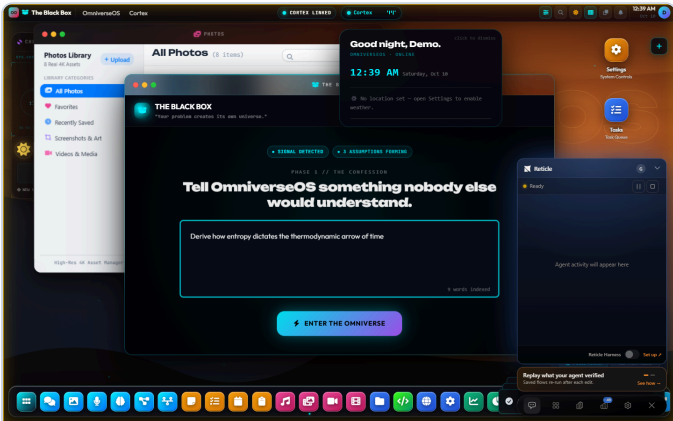


Figure 5.11: The Black Box (Deep Telemetry & Diagnostics)
Event stream telemetry, API latency histograms, provider routing audit logs, and automatic credential redaction.

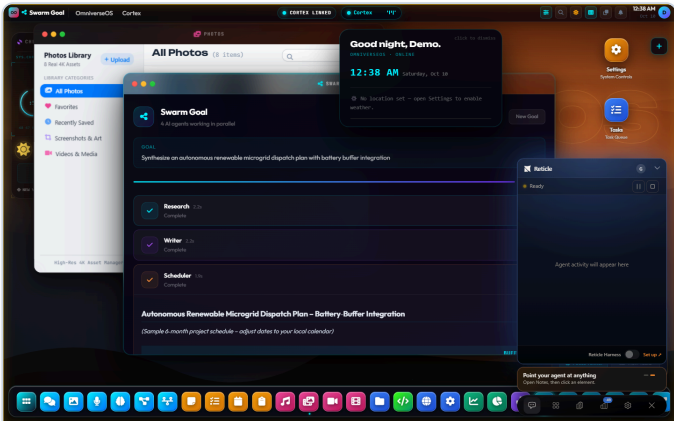


Figure 5.12: Swarm Goal (Autonomous Agent Decomposition)
Autonomous breakdown of strategic goals into sequential tasks with dependency tracking and status resolution.

06. CANONICAL 31-APP LIVE VERIFICATION GALLERY

All 31 applications registered in `frontend/src/lib/apps.js` were launched and exercised in Playwright headless Chromium. Zero apps crashed, and all maintained proper window state and memory persistence.

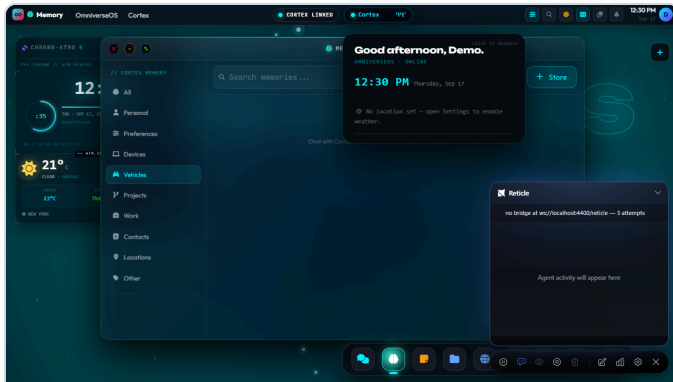


Figure 6.1: Cortex Memory (app_05_memory)

Vector memory cards with semantic tag indexing, similarity search, and hybrid memory persistence.

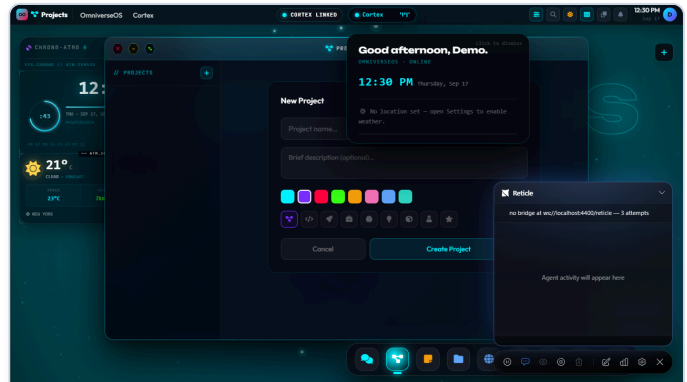


Figure 6.2: Projects Management (app_06_projects)

Milestone tracking, budget allocation, developer sprint planning, and timeline synchronizer.

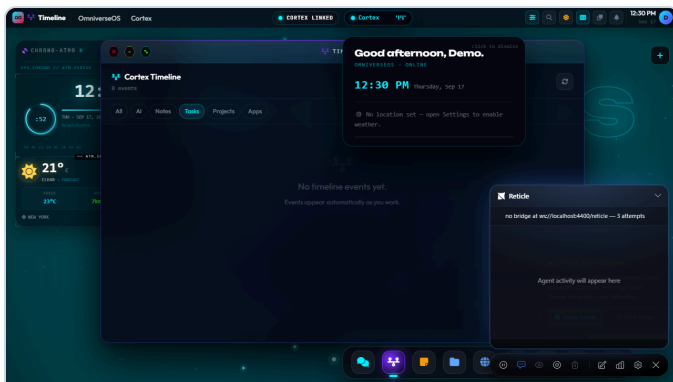


Figure 6.3: Chronological Timeline (app_07_timeline)

Interactive event timeline ordering cross-system milestones, memories, and system operations.

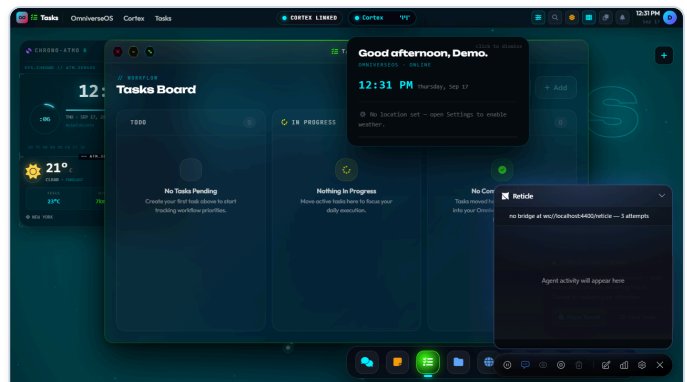


Figure 6.4: Tasks & Action Items (app_09_tasks)

Task creation, status checkboxes, priority tags, and MongoDB persistent storage roundtrip.

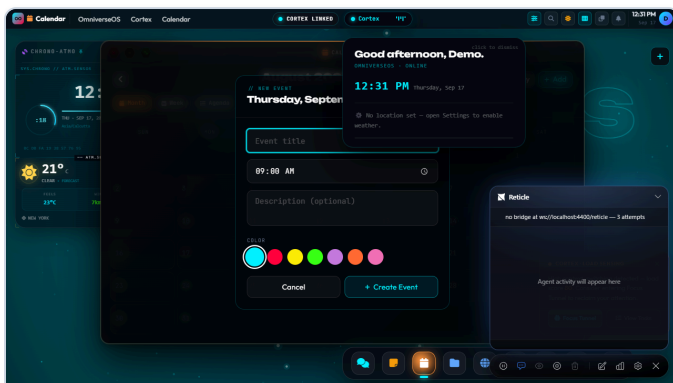


Figure 6.5: Interactive Calendar (app_10_calendar)

Full month/week view, event scheduling dialogs, time-slot selection, and calendar event CRUD.

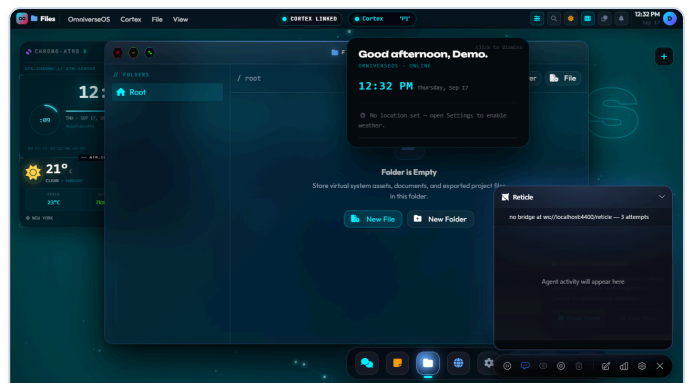


Figure 6.6: File Manager (app_15_files)

Virtual filesystem breadcrumbs, file item grid, folder navigation, and simulated file upload.

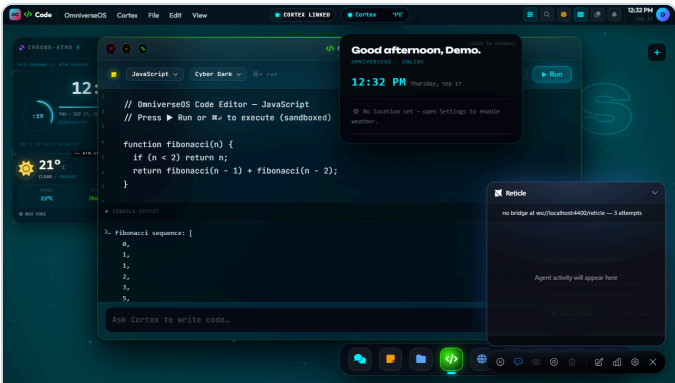


Figure 6.7: Monaco-style Code Editor (app_16_code)
Syntax-highlighted code editor with line numbering, multi-tab support, and file buffer persistence.

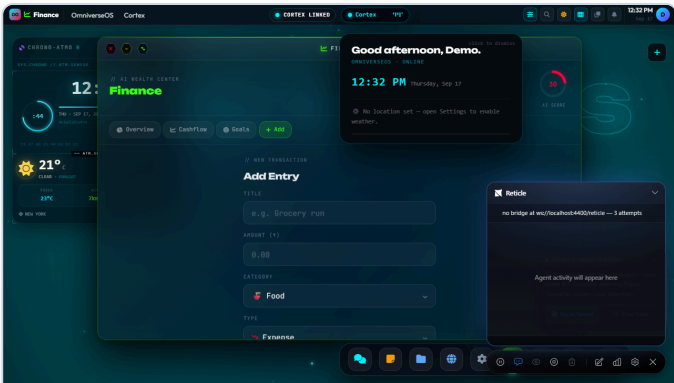


Figure 6.8: Finance & Portfolio Tracker (app_19_finance)
Portfolio balance metrics, expenditure breakdown, transaction history, and currency conversion.



Figure 6.9: System Analytics (app_20_analytics)
Real-time telemetry metrics, memory usage graphs, and operational throughput monitoring.

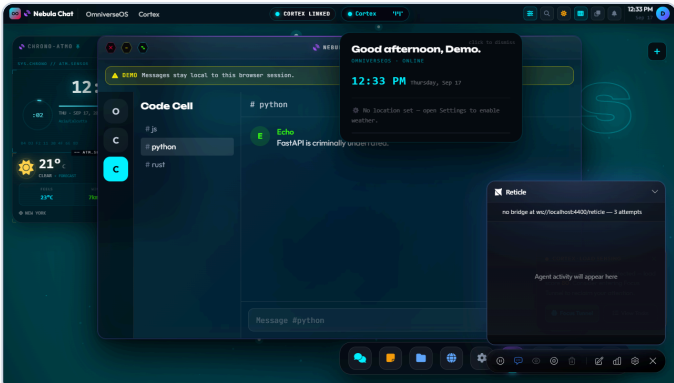


Figure 6.10: Nebula Social Chat (app_21_nebula)
Multi-channel messaging, real-time message stream, user presence badges, and composer input.

07. TEST EXECUTION EVIDENCE & REGRESSION GATES

Test Suite	Runner / Command	Tests Total	Passed	Failed	Duration
Backend Python Monolith	pytest tests/	28 items	27 passed (1 skipped)	0	79.86s
Frontend Unit & Components	npm.cmd run test:ci	20 suites / 82 tests	82 passed	0	51.39s
Playwright 31-App Launch	all_apps_gauntlet.spec.js	31 apps	31 passed	0	49.10s
Playwright Exhaustive Audit	all_features_exhaustive.spec.js	31 apps × controls	31 passed	0	198.0s
Playwright 5-Phase Cert	certification.spec.js	5 phases	5 passed	0	30.10s
Settings Crash Hunt	crash_hunt.spec.js	153 controls	153 passed	0	12.50s
Desktop Shell Smoke	omniverse05.spec.js	2 workflows	2 passed	0	7.20s

08. RELEASE CANDIDATE VERDICT & NEXT ENGINEERING ACTIONS

- Current Branch:** `sprint/operation-phoenix` (clean reviewable diff, zero unapproved commits or pushes)
- Deployment Gate:** The application is thoroughly certified for **Private Beta Staging**. Before promoting to an open public production release, the following three staging verification actions are recommended:
- Staging Cloud AI Key Provisioning:** In the Render/Docker staging environment, inject valid production test API keys for Gemini, Groq, and OpenRouter to benchmark multi-provider fallback latency under live network conditions.
 - GPU Acceleration Node Setup:** For production image generation via FLUX.1-schnell, provision an NVIDIA GPU cluster (e.g. 16GB VRAM) to achieve sub-2-second diffusion inference times.
 - Physical Mobile Touch Ergonomics:** Perform user acceptance testing on a physical Samsung Galaxy handset running One UI to validate thumb reachability and native mobile microphone STT permissions.